

Towards the Development of Transparent Graphene Micro Electrode Arrays

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Introduction and Motivation

Microelectrode arrays (MEAs) [1] are widely used to record the cellular activity. The standard electrode materials in use are platinum, gold and titanium nitride. The entry of carbon based materials in the field of MEAs is only a decade old. Graphene and carbon nanotubes (CNTs) are extensively researched carbon based materials. The application of graphene in optogenetics [2] and calcium imaging [3] is reported in the year 2014. The main advantage of graphene is transparency. Here, we present the fabrication of graphene-based MEAs comprising gold interconnects terminated by graphene electrodes, where an optimal balance between transparency, size of the electrode and impedance is aimed for.

MEA device

Micro Electrode Array (MEA)

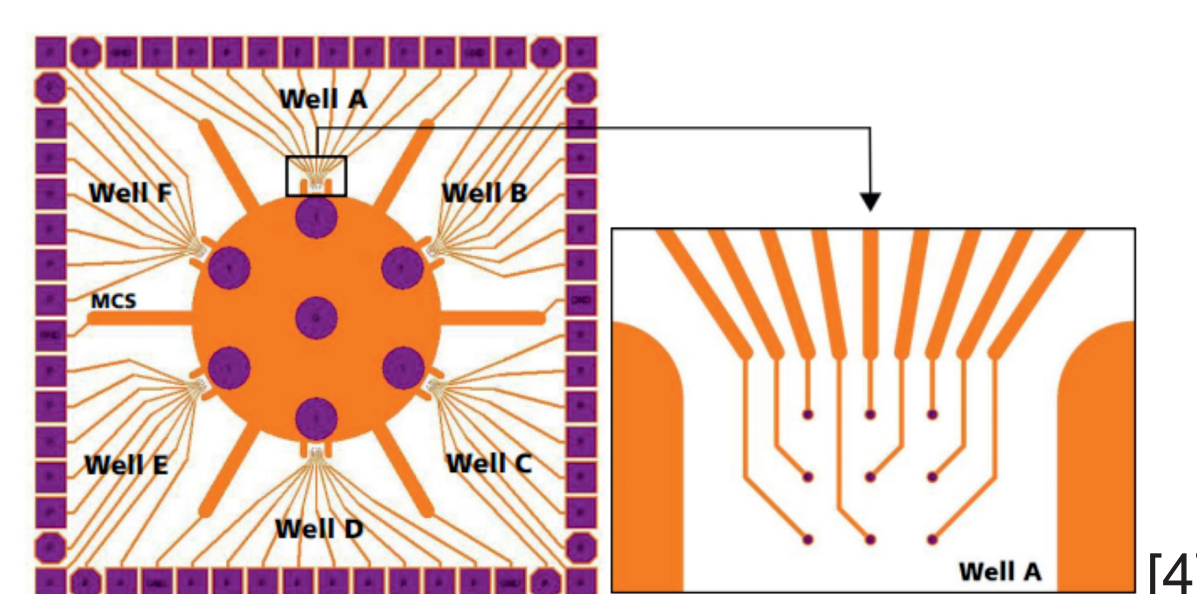


Figure 1: 6-well MEA layout with contact pads and conduction lines. Each well comprises 9 electrodes. The 6-well design offers the possibility to run 6 independent experiments simultaneously.

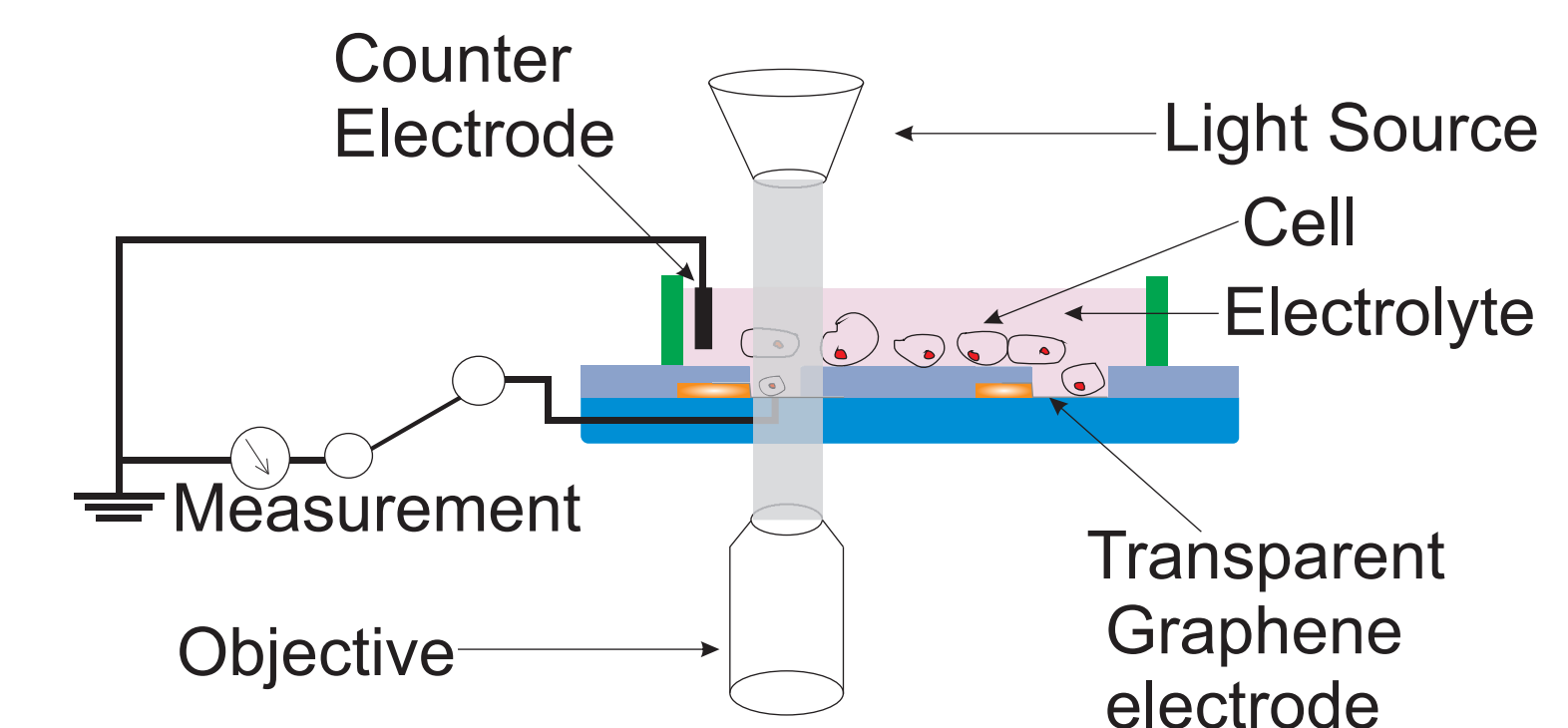


Figure 2: Schematic showing side view of MEA with cell culture on top and objective at the bottom. In the case of conventional electrodes, optical access to the electrode-cell interface is lost due to opaque electrode material. Hence, transparent electrodes need to be developed.

Graphene MEA Fabrication

Graphene processing scheme for integration in MEA

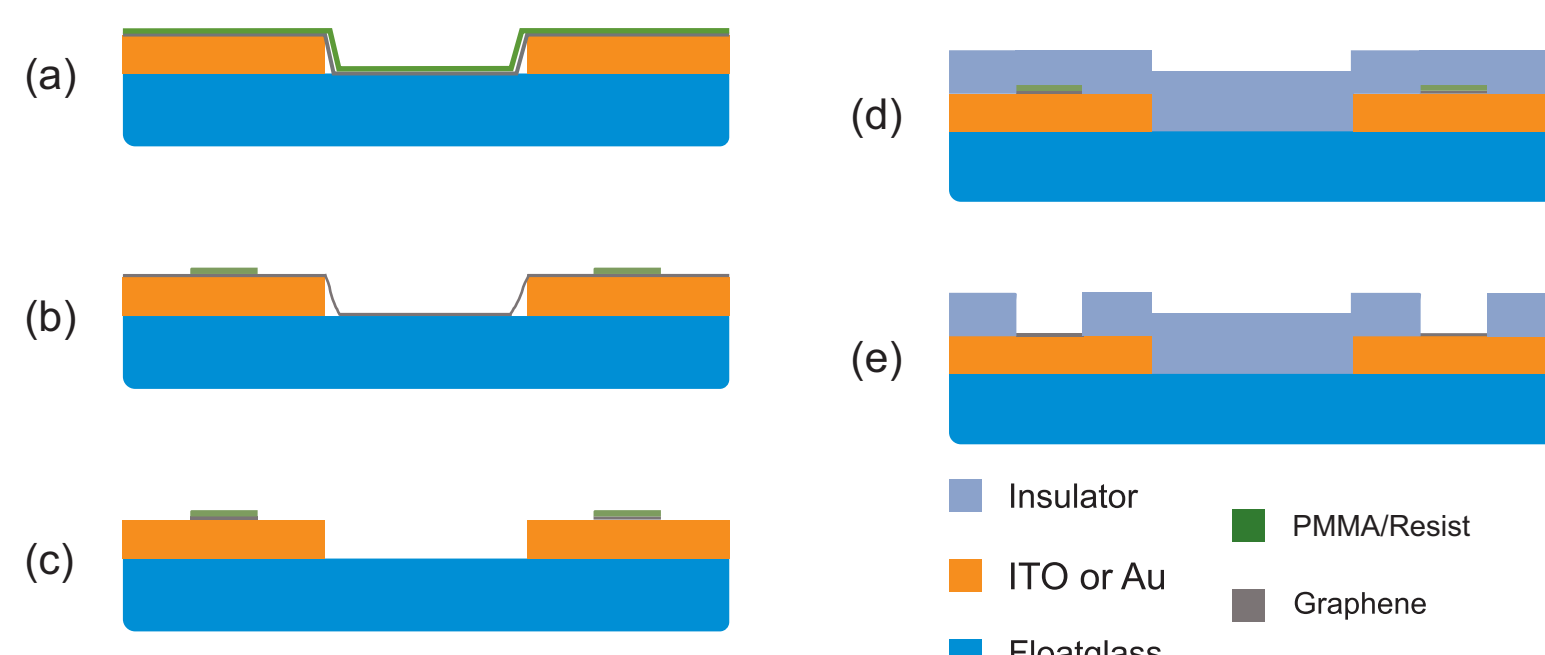
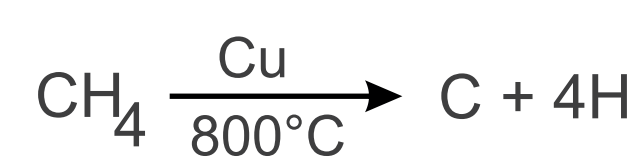


Figure 3: (a) Graphene transfer from Cu to Au conduction lines via PMMA stamp

- (b) PMMA removal followed by photolithography to structure the graphene by oxygen plasma
- (c) Structured graphene with resist
- (d) Insulator over the whole sample
- (e) Opening insulator over the electrodes

Chemical vapour deposition of Graphene



Graphene is produced using methane as carbon source and Cu as catalyst at 800°C at pressures of around 200 mbar. Commercially available polycrystalline Cu foil is pre treated and annealed prior to graphene growth.

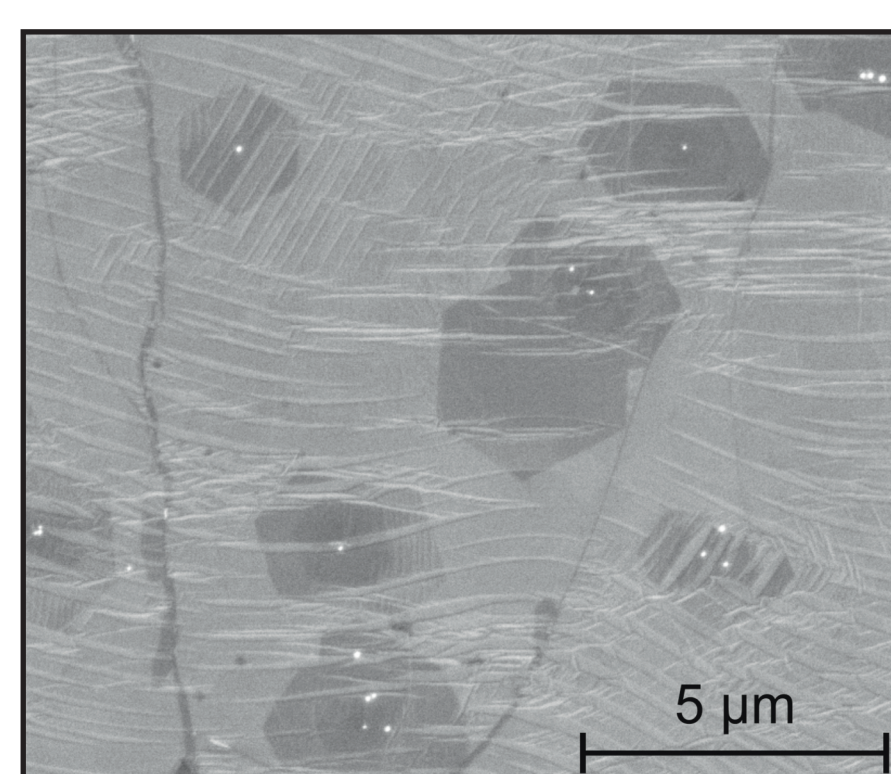


Figure 4: Scanning electron micrograph of Cu with graphene. Bi- and multi-layer graphene is clearly visible.

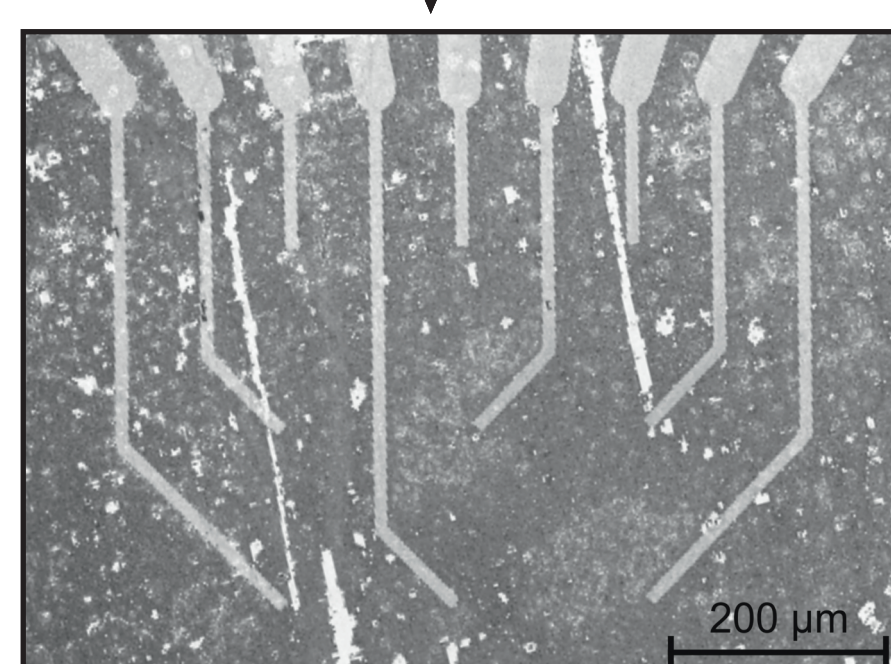


Figure 5: Graphene from Cu foil transferred to Au conduction lines using polymer-based transfer technique.

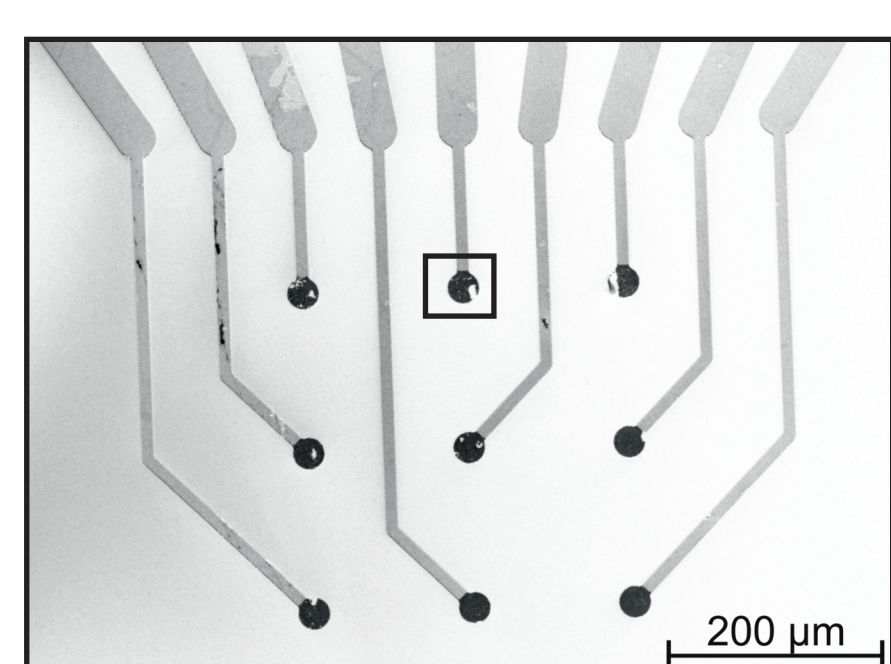


Figure 6: Graphene structured by photolithography and oxygen plasma etching. Resulting in Au conduction lines and largely monolayer graphene electrodes.

Raman Characterization

Raman spectroscopy is widely used to distinguish between different forms of carbon. The graphene Raman spectrum consists mainly of two peaks, namely, G (around 1580 cm⁻¹) and 2D (2680 cm⁻¹). In case of monolayer graphene, the peak intensity ratio 2D/G is two or higher. Furthermore, in the case of bilayer graphene, the 2D/G ratio is less than 2.

The Raman map was obtained with spectra taken with 1 μm spacing. Green laser was used with 1s acquisition time.

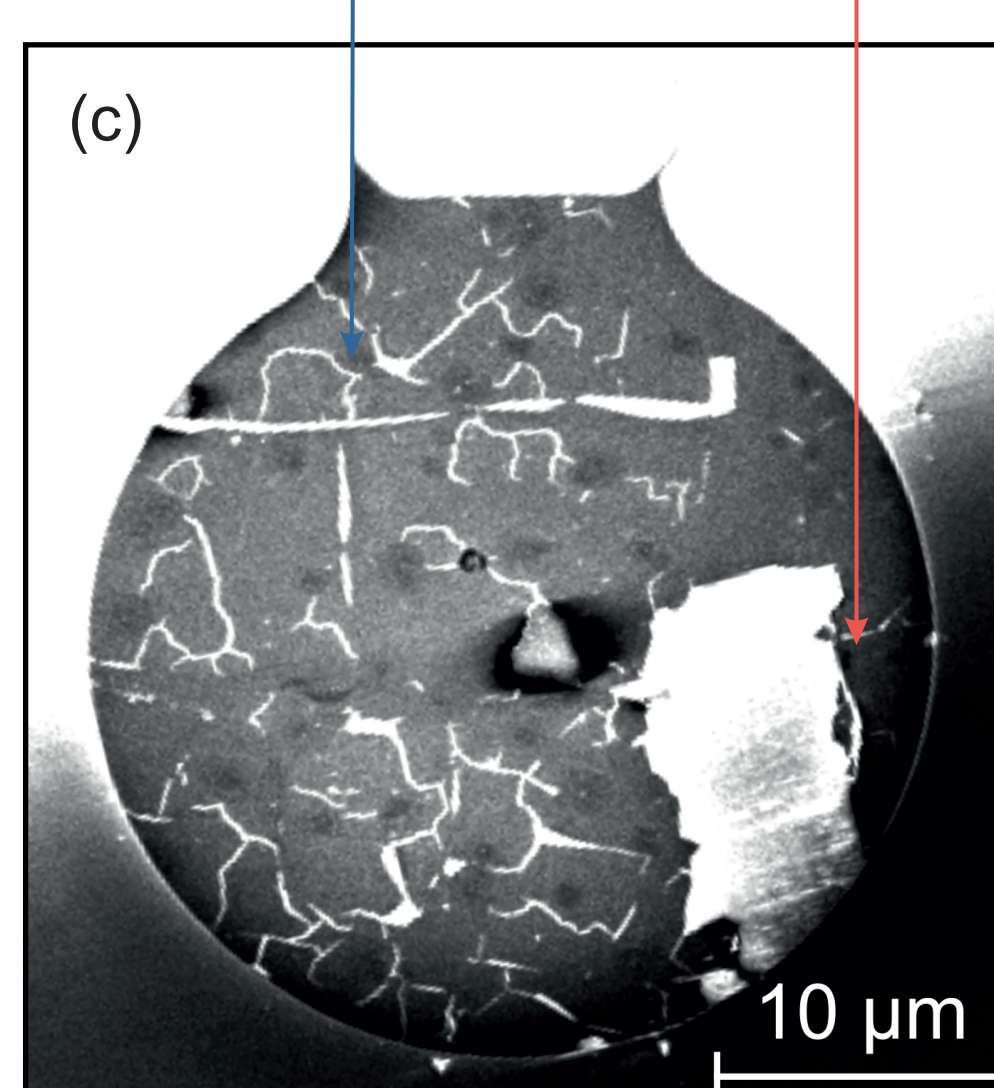
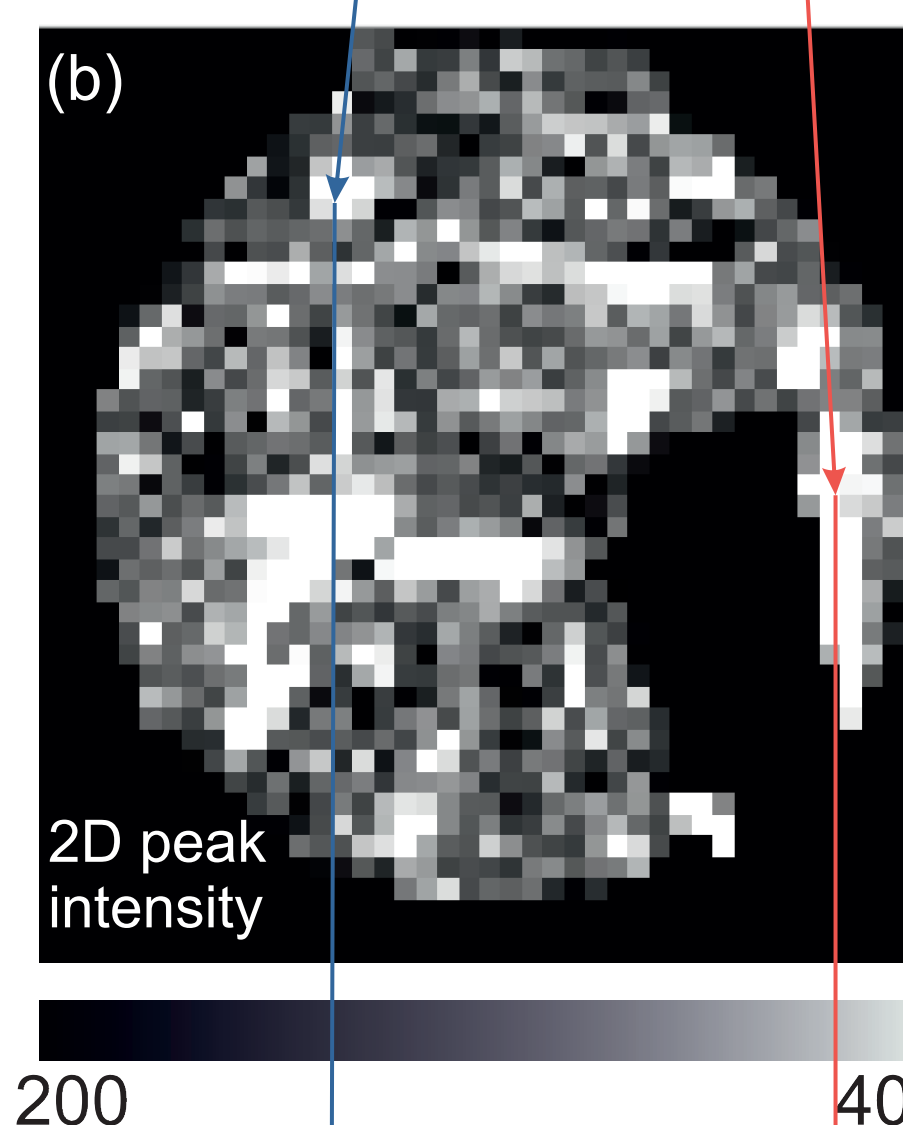
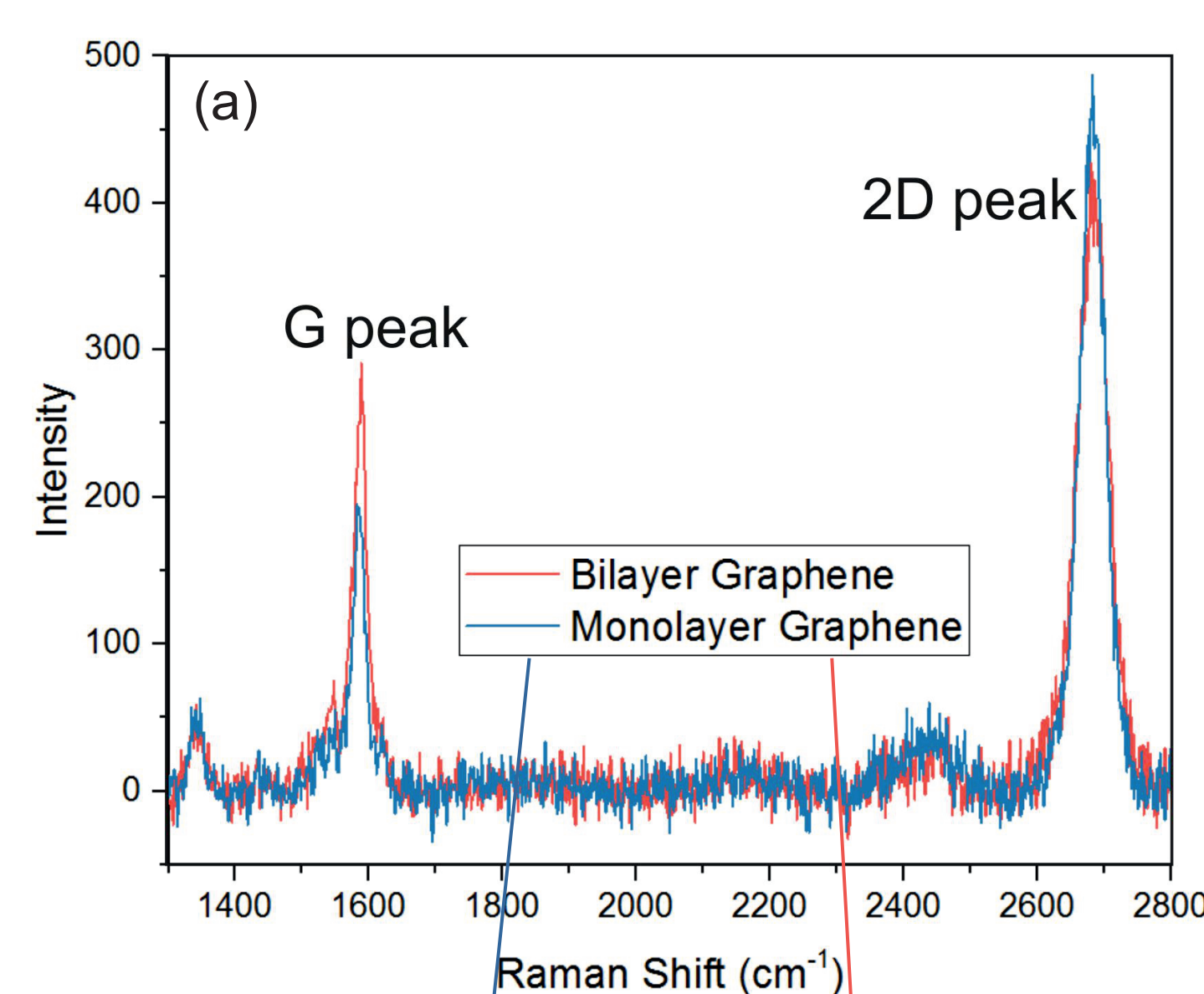


Figure 7: (a) Raman spectra of mono- and bi-layer corresponding to the regions indicated in (b) and (c). (b) 2D intensity map of the electrode in (c). (c) Higher magnification image of electrode indicated in fig 6.

Electrical Characterization

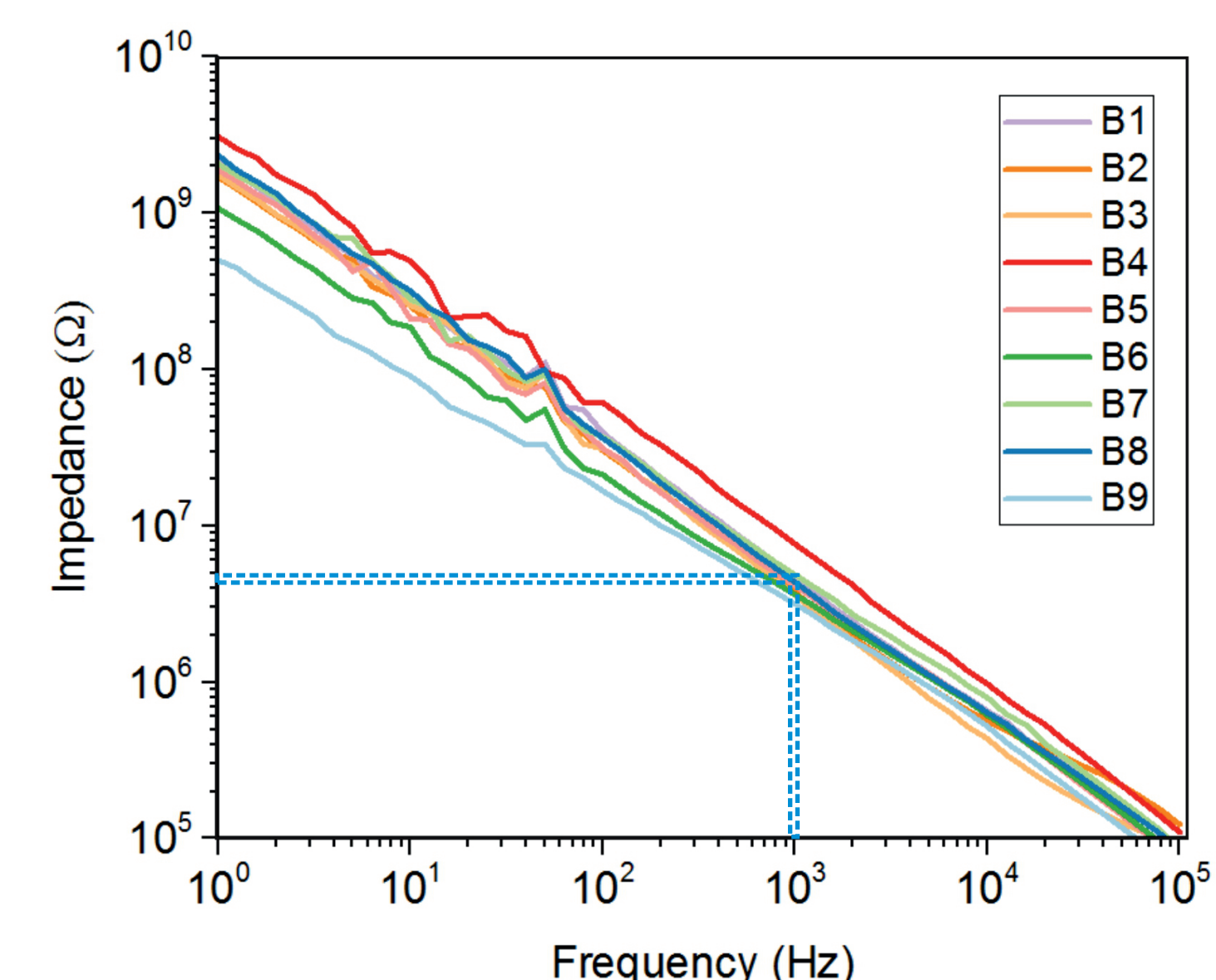


Figure 8: Impedance spectra of 9 graphene electrodes in the well seen in figure 6. The impedance is measured against the internal reference electrode. Measurement is performed with PBS as electrolyte. The impedance value is evaluated at the frequency of 1 kHz.

Results and Outlook

- Reproducible and reliable graphene growth process is established following chemical vapour deposition with Cu as catalyst and methane as carbon source as seen in figure 4.
- Largely monolayer graphene of size 2.5 cm x 2.5 cm is successfully produced and transferred to MEA conduction lines as seen in figure 5.
- Graphene is structured by photolithography (figure 5) and the presence and quality of graphene is checked by SEM imaging and Raman mapping as seen in figure 7. The presence of largely monolayer graphene is confirmed by the Raman spectrum where G and 2D peaks are clearly visible.
- Graphene based transparent MEA is produced and impedance is measured.
- Experiments to test the applicability of the fabricated MEAs for optogenetic measurements and for calcium imaging are planned.

References / Acknowledgements

- [1] A. Stett et al., Anal. Bioanal. Chem. 377:3 (2003)
- [2] D.-W. Park et al., Nat. Comm. 5, 5258 (2014)
- [3] D. Kuzum et al., Nat. Comm., 5, 5259 (2014)
- [4] Microelectrode array (MEA) Manual, Multichannel Systems MCS GmbH

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